Sequence Data and Embeddings

Advanced Machine Learning

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DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Learning objectives

- Explain why sequences differ from independent examples: order matters and length varies
- Name sequences that are not text, and say whether their items are numbers or symbols
- Turn text into tokens and ids, and say what each choice gains and loses
- Encode text as one-hot and bag-of-words vectors, and say what each cannot represent
- Describe an embedding as a learned vector per symbol, and explain how word2vec learns one from neighbours
- Build next-token pairs, and say why a fixed-window language model is limited

Sequences

Order is data

Sequences are everywhere



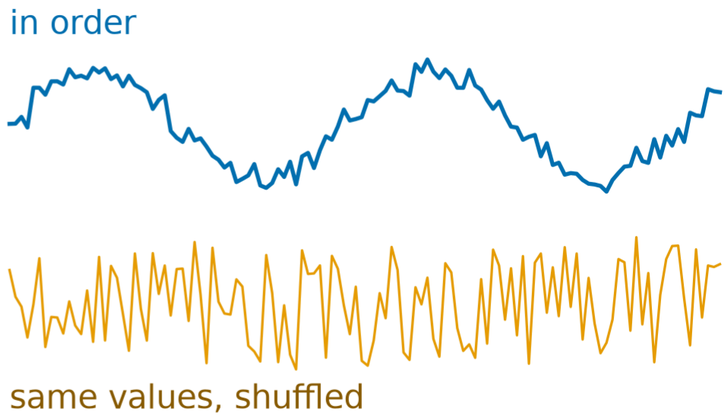
- A **sequence** is a list of items where the position of each one matters
- The items can be words, measurements, letters in a gene, pages a user visited, notes
- This lecture is mostly about words, because that is where the ideas are easiest to see. The same ideas apply to all of these

Order matters

the cat sat on the mat and the cat ran

↓ same items, shuffled

cat the the the cat sat mat ran and on



- The same items in a different order can mean something different, or nothing at all
- Until now an example was a row of features or an image that we could treat on its own. In a sequence, **where** an item sits is part of the example
- Sequences also have different lengths: a sentence of 10 words, a sentence of 40

Two kinds of items

Numbers

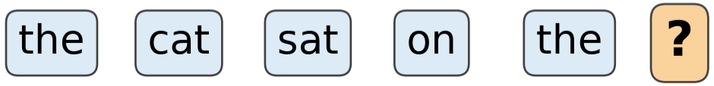
- Sensor readings, prices, audio samples
- **Already numbers**, so they can go into a model directly
- The order is the part we must keep

Symbols

- Words, letters in DNA, pages, notes
- A choice from a list, **not a number**
- We must first decide how to turn each one into numbers

The rest of this lecture is about symbols, and text is our main example

Predict the next item



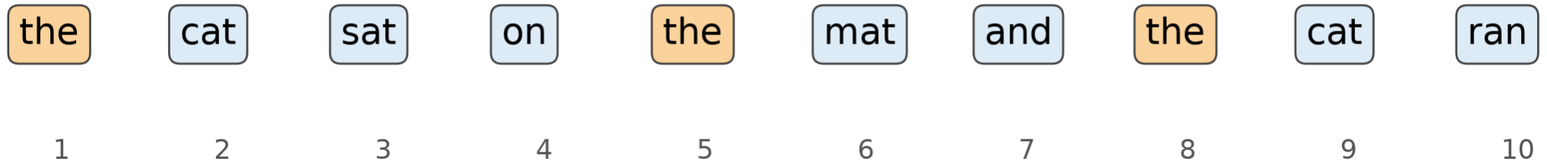


- The simplest sequence task: given the items so far, predict the **next** one
- It fits every example so far: the next word, the next sensor reading, the next note, the next page
- No separate labels are needed. The next item in the data is the answer

The Big Idea

Predict each item from the ones before it, using the same model at every position.

Our running sentence



- 10 items, 7 different words. We trace this one sentence by hand through the whole lecture
- “the” appears 3 times and is followed by cat, mat, cat. To predict what follows it, the words before it matter

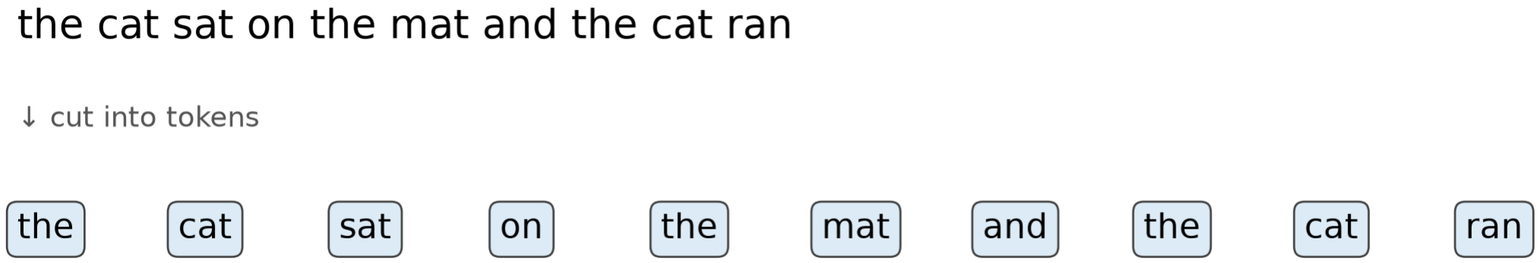
Takeaway

A sequence is data where order matters, and the basic task is to predict the next item from the ones before it.

Text to numbers

A model only sees numbers

From text to tokens



- A **token** is the unit we choose to cut a sequence into
- For text, splitting on spaces is the simplest choice. It is a choice, not the only one
- Symbols in other sequences need the same step: a letter of DNA, or a page visited, is a token

Vocabulary and ids

Token	the	cat	sat	on	mat	and	ran
Id	0	1	2	3	4	5	6
Count	3	2	1	1	1	1	1

- The **vocabulary** is the list of distinct tokens, here 7. Each gets an **id**, its position in the list
- The sentence becomes a list of ids: [0, 1, 2, 3, 0, 4, 5, 0, 1, 6]. An id is only a label: id 1 is not “more” than id 2

Special tokens

Token	What it is for
`<unk>`	Not in the vocabulary: rare or new words
`<pad>`	Filler, so a batch has one length
`<bos>`, `<eos>`	Start and end of a sequence

- The vocabulary is fixed from the training text. Shorter sequences are padded to share a batch
- Names vary (`<s>` and `</s>`, `[CLS]` and `[SEP]`), the roles do not
- From here on our sentence is wrapped: `<bos>` the cat sat on the mat and the cat ran `<eos>`, 12 tokens

What a choice loses

Choice	Before	After	What is lost
Lowercase everything	`The cat` and `the cat`	`the cat`	Which word began a sentence
Drop punctuation	`Stop. Go!`	`stop go`	Where sentences end
Keep only common words	`a quokka sat`	`a <unk> sat`	The rare word itself

- Every preprocessing step throws something away. Whether that matters depends on the task
- Lowercasing is fine for sorting reviews by topic, and not fine for finding names

Characters, words, or pieces of words?

Token	Vocabulary size	Sequence length for our sentence
Characters	12	38
Words	7	10
Pieces of words	in between	in between

- Characters: a tiny vocabulary, but long sequences. Words: short sequences, but a huge vocabulary and new words keep arriving
- Pieces of words (subwords) split a rare word into common parts, such as `un` + `happi` + `ness`. Real systems use these
- The same tradeoff holds for DNA: single letters, or groups of letters called k-mers

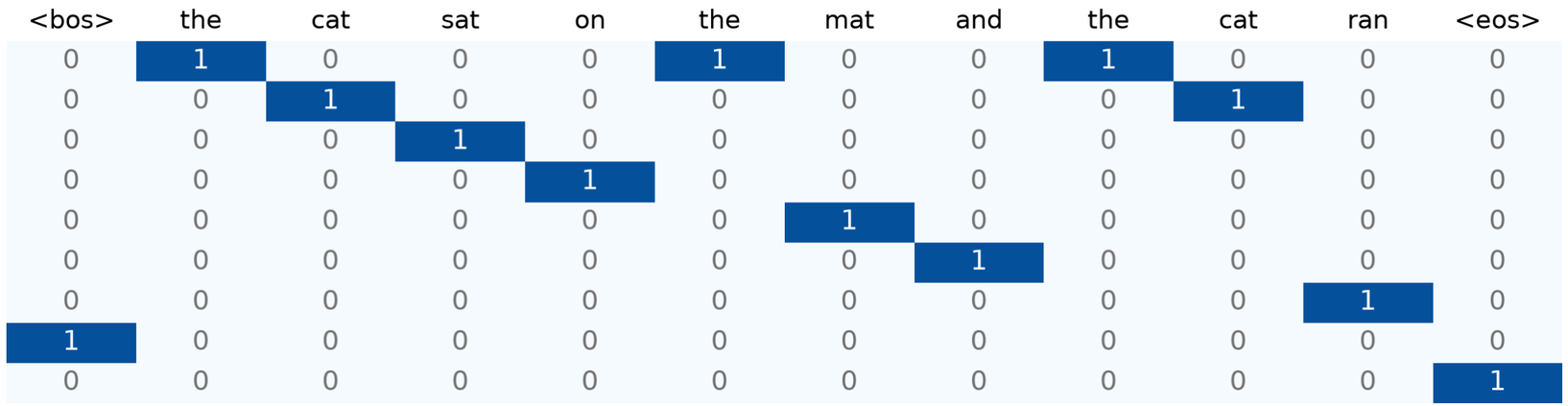
Takeaway

Turning a sequence into tokens decides what the model can see and how long its sequences are.

Encoding words

Simple ways to turn tokens into vectors

One-hot vectors



- A token is one of a fixed list, a category, just like a class label. So we do what we did for class labels: a **one-hot** vector, with a 1 in the slot of the word and 0 elsewhere. Each column is one token of the sentence
- Distance between “cat” and “mat”: 1.41. Between “cat” and “the”: 1.41. Every pair of different words is the same distance apart

Bag of words

Text	the	cat	sat	on	mat	and	ran
the cat sat on the mat	2	1	1	1	1	0	0
the mat sat on the cat	2	1	1	1	1	0	0
the cat ran	1	1	0	0	0	0	1

- A **bag of words** encodes a whole text as one vector of counts, one slot per vocabulary word
- Any length of text gives a vector of the same size
- The first two texts get the **same** vector. The order is gone
- Good for simple jobs such as sorting by topic. A common tweak, TF-IDF, down-weights words that appear in every text

What these cannot say

No sense of similarity

- Every pair of different words is equally far apart
- “cat” is no closer to “mat” than to “the”, and no closer to a word like “kitten”
- A model gets no help from words that behave alike

No sense of context

- One-hot: a word is the same vector wherever it appears
- Bag of words: the order of the words is thrown away
- “the cat sat on the mat” and “the mat sat on the cat” look the same

Both are fine for some jobs, and neither gives the vectors we want: ones where closeness means something

Embeddings

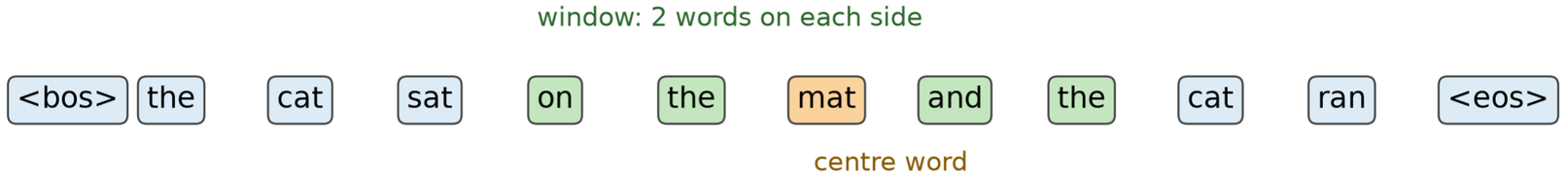
Learn the vectors

The Big Idea

> **“You shall know a word by the company it keeps.”**

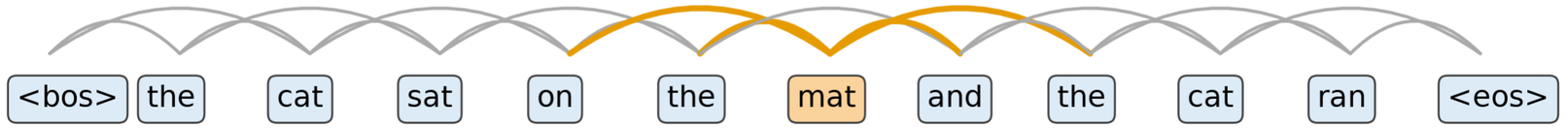
- J. R. Firth, 1957
- A word’s meaning shows in the words that appear around it. That is the idea behind **word2vec**

word2vec: predict the neighbours



- Slide a window along the text. The word in the middle is the **centre word**, the words around it are its **neighbours**
- **Skip-gram**: given the centre word, predict each of its neighbours. The vectors learned along the way are the embeddings

Skip-gram pairs



- Slide the window so every word takes a turn as the centre. Each arc links two words that share a window, giving two pairs: one in each direction
- Orange: the window we just saw. “mat” as the centre gives mat → on, mat → the, mat → and, mat → the
- 42 pairs in all. The labels are free, like the same/different pairs last lecture

A word, in terms of its neighbours

	the	cat	sat	on	mat	and	ran	<bos>	<eos>
the	0	2	2	1	2	2	1	1	0
cat	2	0	1	1	0	1	1	1	1
sat	2	1	0	1	0	0	0	0	0
on	1	1	1	0	1	0	0	0	0
mat	2	0	0	1	0	1	0	0	0
and	2	1	0	0	1	0	0	0	0
ran	1	1	0	0	0	0	0	0	1
<bos>	1	1	0	0	0	0	0	0	0
<eos>	0	1	0	0	0	0	1	0	0

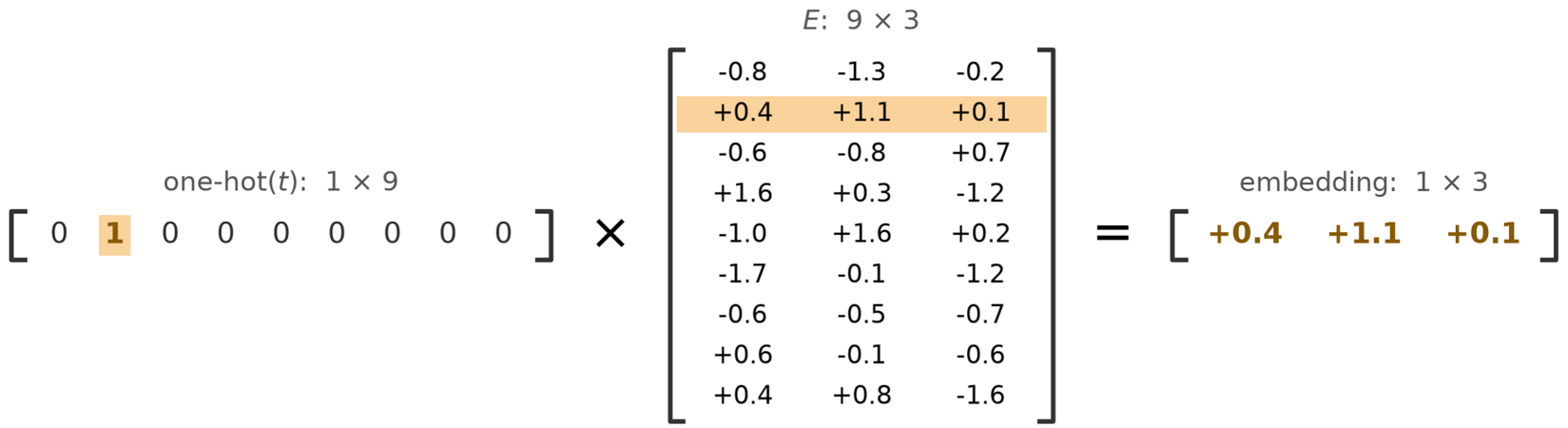
- Count how often each word (row) has each other word (column) within the window: each row is a word **in terms of its neighbours**
- A row is as long as the vocabulary. With thousands of words, it is almost all zeros
- An **embedding** plays that role in a short vector of d numbers

Where the vectors live: the embedding table



- Each token is one row of the **embedding table** E, a matrix of shape (vocabulary size, d), here 9×3. The highlighted row is “cat” (id 1)
- The numbers are parameters. They start random, and training moves them so each row describes its token by its neighbours

Breakdown: a lookup is a matrix product



- The 1 picks out row 1 of E and every other row is multiplied by 0, so the product **is** that row. The layer just looks the row up
- A long, mostly-zero vector in, a short dense one out: 9 numbers become 3. With 20,000 words and $d = 300$, 20,000 become 300

The embedding layer in PyTorch

```python
import torch
from torch import nn

emb = nn.Embedding(num_embeddings=9, embedding_dim=3)       # the table E: 9 tokens, 3 dimensions
ids = torch.tensor([[ID[w] for w in SEQ]])                  # (batch, length) = (1, 12)
out = emb(ids)                                              # (batch, length, dim)
print(ids.shape, "->", out.shape)

onehot = nn.functional.one_hot(ids, 9).float()              # (1, 12, 9)
print(torch.allclose(onehot @ emb.weight, out))             # a lookup equals one-hot times the table
```

```
torch.Size([1, 12]) -> torch.Size([1, 12, 3])
True
```

- Input: ids of shape (batch, length). Output: one vector per id, shape (batch, length, d)
- `emb.weight` is the table, an ordinary parameter that backpropagation updates

How do we train the table?

Inside a task

- Put the table at the front of a model with a task, such as predicting the next word
- That task’s loss is the training signal
- The vectors are a by-product

On neighbours alone

- Train on one question: which words appear near each other?
- word2vec’s own loss is the training signal
- The vectors are the product. This is the route we use

Either way: start from random rows, compute a loss, and backpropagation moves the rows of the table

The skip-gram loss

One training example is a centre word with vector $\mathbf{u}$, one real neighbour $\mathbf{v}$, and a few random words $\mathbf{v_n}$.

$$\text{loss} = \underbrace{-\log \sigma(\mathbf{u} \cdot \mathbf{v})}_{\text{real neighbour}} \underbrace{-\sum_{n} \log \sigma(-\mathbf{u} \cdot \mathbf{v_n})}_{\text{random words}}$$

- It is word2vec's own loss. Training makes it small by scoring the real neighbour high and the random words low
- The same idea as the twin network's contrastive loss, in a different form: real pairs together, other pairs apart

Breakdown: the skip-gram loss

Piece	What it is
$\mathbf{u}$	The **centre** word's vector: its row of the table
$\mathbf{v}$	A **real neighbour**'s vector: a word inside the centre word's window
$\mathbf{v_n}$	A **random** word's vector, drawn from the vocabulary. A few per example, so the sum
$\mathbf{u} \cdot \mathbf{v}$	The score for the pair, read through σ as a probability that they are neighbours
$-\log \sigma(\mathbf{u} \cdot \mathbf{v})$	**Real-neighbour term**: small when the real pair scores high
$-\log \sigma(-\mathbf{u} \cdot \mathbf{v_n})$	**Random-word term**: the minus sign inside flips it, so it is small when the random word scores **low**

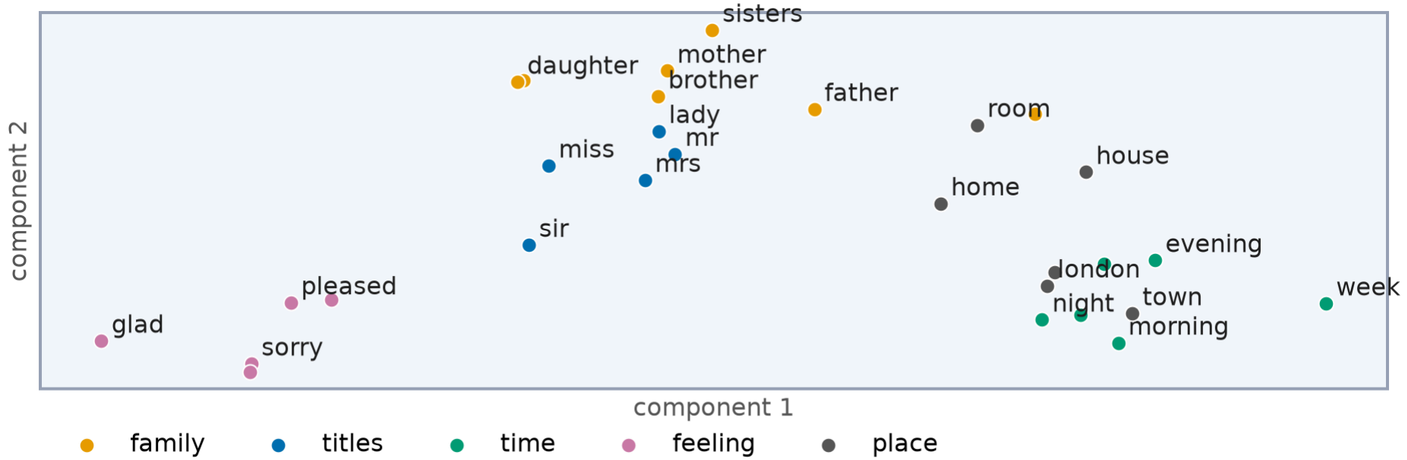
- Without the random-word term, every vector could point the same way and the real-neighbour term would be satisfied
- How it is used: for each (centre, neighbour) pair, raise the score of the two rows, and lower the centre's score with a few random words (**negative samples**). Backpropagation moves the rows

The skip-gram loss with real numbers

Pair	Dot product	The term with the number in	Loss
Real neighbour, score high	$+2$	$-\log \sigma(2)$	0.13
Real neighbour, score low	-2	$-\log \sigma(-2)$	2.13
Random word, score low	-2	$-\log \sigma(2)$	0.13
Random word, score high	$+2$	$-\log \sigma(-2)$	2.13

- Because $\sigma(2) = 0.88$: a real pair that scores $+2$ costs 0.13, and one that scores -2 costs 2.13
- For a random word the signs flip. Scoring high costs 2.13, so training lowers that score

What the vectors look like



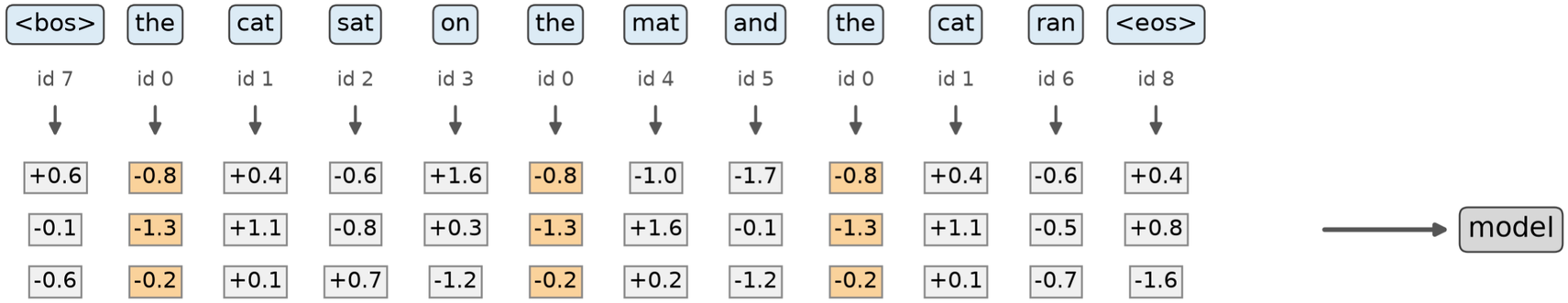
- word2vec (50 dimensions) trained on 128,565 words of *Pride and Prejudice*. Shown: a 2-D PCA of 29 chosen words, colored by group (only some labeled)
- Nobody told it the groups. Words used in similar company ended up near each other. Two dimensions keep only 28% of the variance, so it is a rough view

Nearest neighbours

Word	Its nearest neighbours
mother	father, sister, commendation, husband, song
london	town, meryton, scotland, hertfordshire, militia
mr	collins, miss, wickham, steward, darcy
morning	day, saturday, next, third, arrival
money	knows, exertion, respect, done, pardon

- Nearest by cosine similarity of the vectors. Family words, places and titles come out sensible
- “Near” means “used in similar company”, not “means the same”. Rare words and a small book give some odd neighbours, such as for “money”

A sentence becomes a sequence of vectors



- Look up each token's row: the sentence becomes a **sequence of vectors**, shape (12, 3). This is what the model is given
- The order is kept, and a repeated word gets the same vector: "the" (orange) three times
- A batch has shape (batch, length, d), as in the PyTorch slide

One word, one vector

“bank”

- “she sat on the river **bank**”
- “she went to the **bank** for a loan”

One vector

- Every encoding so far gives “bank” the same vector in both sentences
- The vector cannot depend on the words around it

Making a word’s vector depend on its context is where the next lecture, and then attention, go

Takeaway

An embedding layer turns ids into vectors, and word2vec trains them so that words used in similar company end up close.

Language modeling

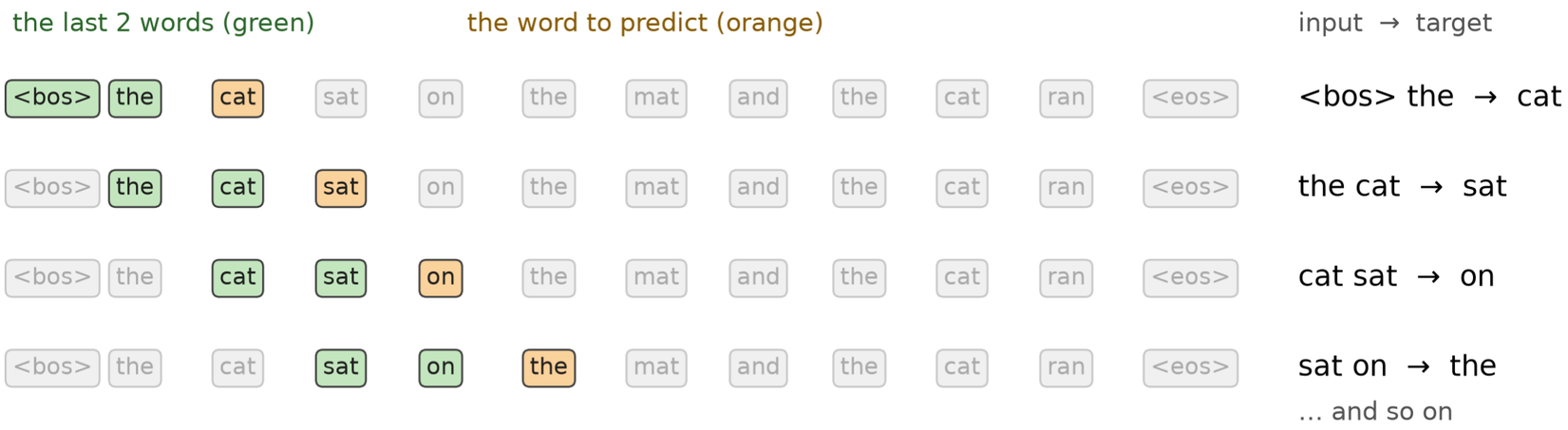
Predicting what comes next

Predicting the next word

Words are now vectors. What do we ask a model to do with them?

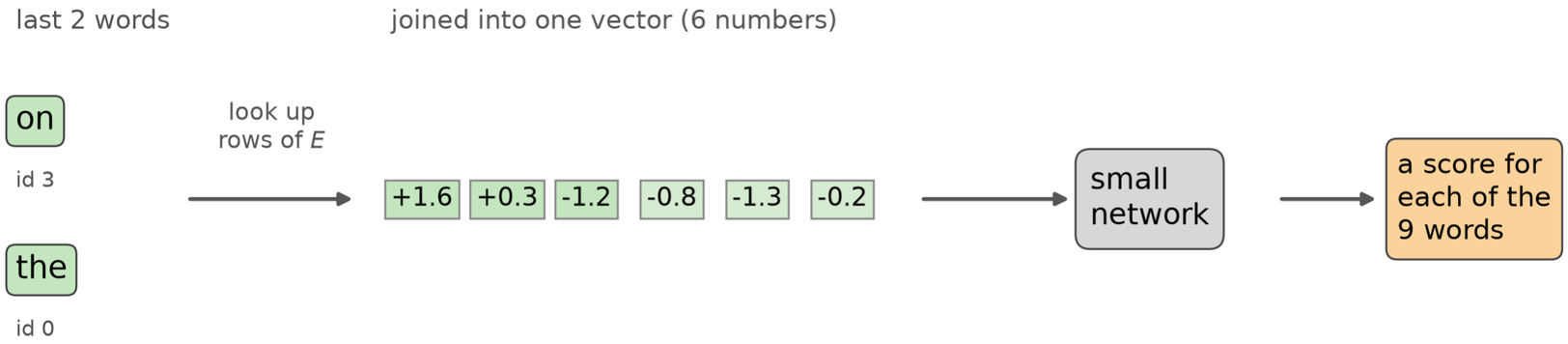
- The task from the start of the lecture: given the words so far, predict the next one. “the cat sat on the ___”
- It is a **classification** problem. The input is the words so far, the classes are all the words in the vocabulary, and the correct class is the word that actually came next
- The labels cost nothing: any text already contains the right answer at every position
- A model that does this is a **language model**

Next-token pairs



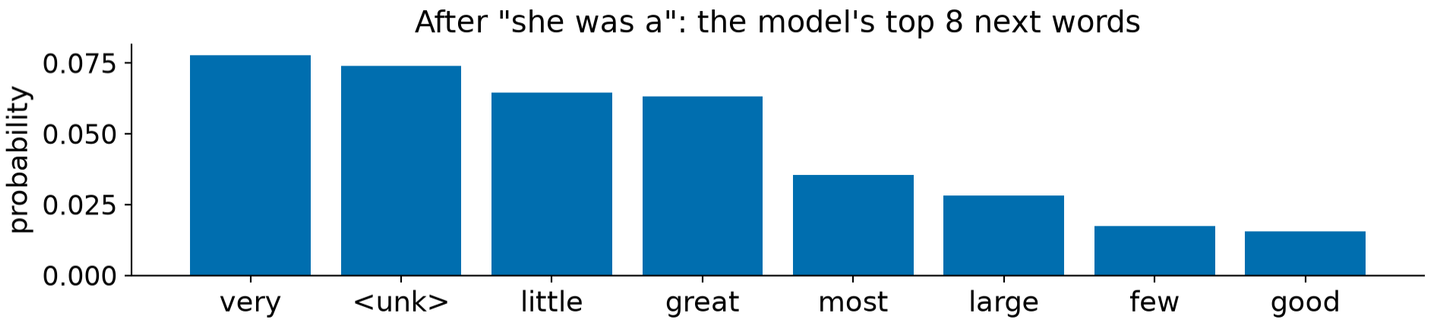
- Slide a window of the last 2 words along the sentence. Each position gives one training example: the window is the input, and the word after it is the target
- 10 examples from 12 tokens, and every one comes straight from the text. Skip-gram used neighbours on both sides. Here only the past counts

A fixed window of context



- Look up the embeddings of the last k words and feed them to a small network that scores every word. Here we join them into one vector of $k \times d$ numbers. Other models combine them differently
- The same network is used at every position, as the Big Idea said. It works when a little context is enough
- Training: backpropagation updates the network and the table rows it used

The model’s output



- Trained on the whole novel. One score per word (2,124 of them), turned into probabilities by softmax
- Embeddings go in, but the output is not an embedding: it is a score per word id, trained with cross-entropy against the true next word (a one-hot). Classification with 2,124 classes
- The probabilities spread over plausible words, as they should: no model knows which one the author wrote

Breakdown: softmax over the vocabulary

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \qquad \text{loss} = -\log p_{\text{true}}$$

Piece	What it does	After “the”
z_i	Score for word i	cat 2, mat 1, ran -1
e^{z_i} and $\sum_j e^{z_j}$	Make positive, divide by the total	7.39, 2.72, 0.37; total 10.48
p_i	Probability of word i	0.71, 0.26, 0.04
$-\log p_{\text{true}}$	Small if the true word got a high share	cat: 0.35, ran: 3.35

- Same softmax and cross-entropy as classification, with all the words as the classes

The limit of a fixed window

Window	Contexts followed by more than one word in our sentence
1 word	“the” → cat, mat; “cat” → ran, sat
2 words	“the cat” → ran, sat
3 words	none

- With 1 word we cannot tell what follows “the” or “cat”. With 2 words, “the cat” is still followed by both sat and ran. It takes 3 words here to tell them apart
- Real text always has something that matters from further back than any window we pick: the subject of a long sentence, a name from a paragraph ago
- A longer window means a bigger input, and everything older than the window is invisible

What a longer window costs

Window k	Input size	Held-out perplexity
none (word frequencies only)	0	343
1	32	148
2	64	138
3	96	140
5	160	143
8	256	150

- Perplexity is e to the held-out cross-entropy. Lower is better: roughly how many equally likely words the model chooses between
- One word of context helps a lot. Past 2 words, on this much text, a longer window does not

Same table, different training signal

Word	Trained on neighbours (word2vec)	Trained inside a next-word model
mother	father, sister, commendation, husband	arose, stand, reason, child
london	town, meryton, scotland, hertfordshire	eliza, rejoice, choose, while
mr	collins, miss, wickham, steward	mrs, drawn, favourite, chiefly

- The window model's embedding table is trained inside a task, the first route from earlier. word2vec's is trained on neighbours alone
- Same text, same kind of table: the training signal decides what the vectors reward
- Trained only to predict the next word on this little text, they are far less tidy. word2vec asks for neighbours directly

Takeaway

A language model is a next-word classifier over the whole vocabulary, and a fixed window sees only a fixed amount of the past.

Summary

Recap: ways to encode a token

	Vector size	Similar words close?	Built by
One-hot	vocabulary size	No: all equally far	By hand
Bag of words	vocabulary size	No, and word order is lost	By counting
Embedding inside a task	d, small	Only as far as the task needs	Training on the task's loss
word2vec	d, small	Yes, for words used in similar company	Training on neighbours

- All four give a word the **same** vector wherever it appears

Recap: key ideas

- A **sequence** is data where order matters, and the basic task is to predict the next item. Items can be numbers, or symbols such as words, letters in DNA, pages, notes
- Symbols become **tokens**, then **ids**. Every choice (case, punctuation, characters or words) throws something away
- **One-hot** and **bag of words** are simple encodings with no sense of similarity or order
- An **embedding** is a learned vector per token. A lookup is a one-hot vector times a table. **word2vec** trains it with the skip-gram loss on neighbours
- A **language model** predicts the next word. A fixed window sees only a fixed amount of the past

Where this goes next

- A model that carries a summary of everything seen so far, and updates it with each new word: a recurrent network
- Then attention: a vector that depends on the words around it, so “bank” can differ
- Not covered here: how subwords are built, seq2seq, and pretrained embeddings